

Laporan Minggu 4

Monitoring, Keamanan, dan Optimasi
Final Project Cloud Computing - Kelompok 11
Cloud-Based Data Processing and Monitoring Platform on Microsoft Azure

Nama	Peran
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Tujuan

Minggu 4 berfokus pada operasional sistem: monitoring, alerting, centralized logging, security audit, backup/recovery, dan optimasi biaya.

Monitoring dan Alerting

Area	Implementasi	Tujuan
Application Insights	Terhubung ke Azure Functions	Request log, latency, exception
Azure Monitor	Alert rule operasional	Deteksi error dan performa
Action Group	Email notifikasi tim	Respons incident
Dashboard aplikasi	Statistik, tabel, dan chart	Monitoring data hasil proses

Alert Rule Yang Disiapkan

Alert	Target	Threshold
Function 5xx Error	Azure Function App	HTTP 5xx > 5 dalam 5 menit
Function Latency	Azure Function App	Average response time > 2 detik
VM CPU High	VM Web	CPU rata-rata > 80 persen selama 5 menit

Security Baseline

Area	Kontrol	Status
Frontend secret	Browser hanya memanggil /api	Selesai
Function key	Cloudflare env server-side	Selesai
Cosmos secret	Azure Key Vault	Selesai
Password	PBKDF2 hash	Selesai
Role	Register publik role user	Selesai
Data isolation	Filter owner_user_id	Selesai
Repo hygiene	.env/.tfvars/state/key di-ignore	Selesai
VM SSH	Batasi IP admin	Perlu hardening

Backup dan Recovery

- Cosmos DB mengandalkan backup platform Azure sesuai konfigurasi account.
- Blob Storage raw-data dapat digunakan untuk reprocessing.
- Source code tersimpan di GitHub sebagai version control.
- Simulasi recovery disarankan: upload file kecil, verifikasi Cosmos DB, export sample record, lalu proses ulang file.

Cost Optimization

Optimasi	Dampak
Cloudflare Pages	Mengurangi kebutuhan VM untuk frontend
Azure Functions Consumption	Mengurangi biaya backend idle
Cosmos DB Serverless	Cocok untuk traffic kecil proyek

Optimasi	Dampak
Storage lifecycle policy	Mengontrol pertumbuhan file mentah
Matikan VM saat tidak demo	Mengurangi biaya compute

Screenshot Yang Perlu Diambil Dari Portal

Bukti	Lokasi
Application Insights overview	Azure Portal > Application Insights
Alert rules dan action group	Azure Monitor > Alerts
Cost breakdown	Azure Cost Management
Security posture	Defender for Cloud
Backup/recovery evidence	Cosmos DB / Storage

Catatan

Sebelum screenshot portal dimasukkan ke laporan, pastikan tidak ada function key, Cloudflare API token, connection string, access key, password, file tfvars, atau tenant/subscription ID penuh yang terlihat.